Chenmu Zhang

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SUMMARY

Computational materials scientist (Ph.D., UT Austin; postdoc, Rice) developing, applying and evaluating autonomous LLM agents for scientific research.

Developed MatClaw, an open-source LLM agent for data analysis and long-running HPC simulations.

Applied agents to materials discovery and autonomous model development, including a research loop that improved band-gap prediction through model redesign, training and evaluation with no human in the loop.

Built CompMat-Bench, a benchmark of 94 tasks from published materials studies, evaluating agents on individual research steps and multi-step workflows against reproduced results.

RESEARCH EXPERIENCE

► **Postdoctoral Research Associate** May 2023 – Present

Rice University, Houston, TX Advisor: Prof. Boris I. Yakobson

- Led the design and agent evaluations for **CompMat-Bench**, measuring performance on individual research tasks and multi-step workflows across three LLMs using rule-based grading against reproduced results.
- Designed and built **MatClaw**, an open-source LLM agent for data analysis and long-running materials simulations on high-performance computing (HPC) clusters.
- Applied an autonomous research loop to band-gap model development: the agent redesigned, trained and evaluated models over **184 experiments with no human in the loop**. The resulting model outperformed 17 published expert-designed models trained without external pretraining, based on their reported MatBench scores (LM4Sci workshop, COLM 2026).
- Applied MatClaw to a materials-discovery study, connecting generative structure prediction, interatomic-potential screening and DFT validation in one simulation campaign (co-first-author manuscript in preparation).

► **Graduate Research Assistant** Aug. 2018 – Apr. 2023

University of Texas at Austin, Austin, TX Advisor: Prof. Yuanyue Liu

- Used first-principles calculations to explain transport limitations and identify high-mobility 2D semiconductors; extended the work to substrate and surface effects. Publications include Physical Review Letters (2020, 2023), ACS Nano (2024) and Nano Letters (2026).

SELECTED PROJECTS

► **CompMat-Bench | Agent benchmark and evaluation data** [Code] [Paper]
Released code, task data and **976 agent trajectories** covering **94 tasks** and 10 workflows from 15 published studies. Agents prepare simulation inputs and analyze precomputed outputs, enabling repeatable evaluation without rerunning expensive simulations. Fixed-rule grading avoids reliance on an LLM judge.

► **MatClaw | Autonomous scientific agent** [Code] [Paper]
Sole developer of an agent that writes and executes Python, creates its own tools, and submits and monitors remote HPC jobs. A persistent Python session preserves computational state across steps, supporting data analysis and hours-long simulation workflows.

► **Autonomous LLM Research for Materials Property Prediction** [Code] [Paper]
Built a closed research loop in which an LLM agent modifies a crystal graph neural network, trains and evaluates it, and uses experiment results to guide subsequent changes. Applied to band-gap prediction on MatBench; code publicly available.

TECHNICAL SKILLS

Agent evaluation: benchmark design; rule-based grading; multi-step workflow evaluation; pass-rate, cost and reliability analysis.

Agent systems: Python code execution; tool-use agents; long-horizon workflows; HPC job orchestration; integration with scientific simulation tools.

Scientific computing: DFT (VASP, Quantum ESPRESSO); molecular dynamics (LAMMPS); materials workflows (pymatgen, ASE, atomate2, jobflow); application of interatomic potentials and generative models (MACE, MatterSim, MatterGen).

SELECTED PUBLICATIONS

13 peer-reviewed publications (8 first-author), 3 under review, including *Phys. Rev. Lett.*, *J. Am. Chem. Soc.*, *ACS Nano*, and *Nano Lett.*; 613 citations, h-index 10 (Google Scholar, Aug. 2026). Full list: Google Scholar.

1. **Zhang, C.**; Yakobson, B. I. MatClaw: An Autonomous Code-First LLM Agent for End-to-End Materials Exploration. arXiv preprint arXiv:2604.02688, **2026** [Paper].
2. **Zhang, C.**; Felix, L. C.; Zhang, J.-J.; Li, X.; Jiang, X.; Jiang, T.; Mishra, S.; Qin, X.; Yakobson, B. I. CompMat-Bench: Benchmarking AI Agents for Computational Materials Science. arXiv preprint arXiv:2610.00636, (Submitted) **2026** [Paper].
3. **Zhang, C.**; Yakobson, B. I. Optimizing Expert-Designed Crystal Graph Networks for Band-Gap Prediction with an Autonomous LLM Research Loop. arXiv preprint arXiv:2606.29717, accepted at the Workshop on Language Models for Scientific Discovery (LM4Sci), COLM **2026** [Paper].
4. Yuan*, Z.; **Zhang*, C.** Off-Stoichiometric Energetic and Electronic Asymmetry in Ternary Cu–In–Se: Task-Oriented Materials Discovery by an Autonomous LLM Agent. (In preparation; * equal contribution) **2026**.
5. **Zhang, C.**; Liu, Y. Enhancing Carrier Mobility by Remote Phonons. *Nano Letters* **2026**, 26(30), 9716–9721 [Paper].
6. **Zhang, C.**; Liu, Y. Electron-Surface Scattering from First-Principles. *ACS Nano* **2024**, 18(40), 27433–27439 [Paper].
7. **Zhang, C.**; Wang, R.; Mishra, H.; Liu, Y. Two-Dimensional Semiconductors with High Intrinsic Carrier Mobility at Room Temperature. *Physical Review Letters* **2023**, 130(8), 087001 [Paper].
8. Cheng, L.; **Zhang, C.**; Liu, Y. Why Two-Dimensional Semiconductors Generally Have Low Electron Mobility. *Physical Review Letters* **2020**, 125(17), 177701 [Paper].

SELECTED PRESENTATIONS

- How Well Can AI Agents Do Computational Materials Research? A Benchmark Built from Peer-Reviewed Publications. MRS Fall Meeting, Symposium MT01, Boston, Dec. 2026 (talk, scheduled).
- Optimizing Expert-Designed Crystal Graph Networks for Band-Gap Prediction with an Autonomous LLM Research Loop. LM4Sci Workshop at COLM 2026, San Francisco, Oct. 2026 (poster).

EDUCATION

University of Texas at Austin, Austin, TX Aug. 2018 – Apr. 2023

Ph.D. in Materials Engineering
Advisor: Prof. Yuanyue Liu
Thesis: *"Phonon-limited Carrier Transport in 2D Materials from First Principles"*

University of Science and Technology of China (USTC), Hefei, China Sep. 2014 – Jul. 2018

B.S. in Physics
Advisor: Prof. Wenguang Zhu
Thesis: *"First Principles Calculations on 2D Ferroelectric Materials under Uniaxial Strain"*